

Exact Asymptotic of the Lower Bound

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1.1 Derivation of the Recurrence

Recall that our shifted yield recurrence is

$$Y_{n+1} \stackrel{d}{=} I^{(n)} \left(Y_{U_n}^{(1)} + Y_{n-1-U_n}^{(2)} + 1 \right) + J^{(n)} \max \left(Y_{U_n}^{(1)}, Y_{n-1-U_n}^{(2)} \right), \quad Y_1 = Y_2 = 0.$$

Taking expectation on both sides, and letting $y_n = \mathbb{E}[Y_n]$, we have

$$y_{n+1} = \frac{2u}{n} \sum_{i=0}^{n-1} y_i + \frac{v}{n} \sum_{i=0}^{n-1} \mathbb{E} \left[\max \left(Y_i^{(1)}, Y_{n-1-i}^{(2)} \right) \right] + u, \quad n \geq 1, \quad y_0 = y_1 = 0.$$

Since $\max(Y_i, Y_{n-1-i}) \geq Y_i$ and $\max(Y_i, Y_{n-1-i}) \geq Y_{n-1-i}$, we have

$$\mathbb{E}[\max(Y_i, Y_{n-1-i})] \geq \max(y_i, y_{n-1-i}).$$

Hence, for every $n \geq 1$, we have

$$y_{n+1} \geq \frac{2u}{n} \sum_{i=0}^{n-1} y_i + \frac{v}{n} \sum_{i=0}^{n-1} \max(y_i, y_{n-1-i}) + u.$$

Define (y'_n) such that $y'_0 = y'_1 = 0$ and for every $n \geq 1$,

$$y'_{n+1} = \frac{2u}{n} \sum_{i=0}^{n-1} y'_i + \frac{v}{n} \sum_{i=0}^{n-1} \max(y'_i, y'_{n-1-i}) + u.$$

Then $y_n \geq y'_n$ for every $n \geq 0$. Finally, we homogenize the recurrence by defining

$$x_n = \frac{2u + v - 1}{u} y'_n + 1$$

for every $n \geq 0$. We have $x_0 = x_1 = 1$ and for every $n \geq 1$. The recurrence of (x_n) is

$$x_{n+1} = \frac{2u}{n} \sum_{i=0}^{n-1} x_i + \frac{v}{n} \sum_{i=0}^{n-1} \max(x_i, x_{n-1-i}), \quad n \geq 1, \quad x_0 = x_1 = 1. \quad (1.1)$$

We consider the case where $u + v > 1/2$, which is equivalent to $2u + v > 1$.

1.2 Analysis of the Recurrence

Throughout this section, let α be the unique solution of the equation

$$\alpha + v2^{-\alpha} = 2(u + v) - 1. \quad (1.2)$$

Note that $\alpha > 0$ since otherwise, we would have $2u + v = 1$.

Proposition 1. *The sequence (x_n) is increasing.*

Proof. We have $x_0 = x_1 = 1$ and $x_2 = 2u + v$. Assume that $x_{i+1} \geq x_i$ for all $0 \leq i \leq n-1$ with $n \geq 2$. We have

$$\begin{aligned} nx_{n+1} &= 2u \sum_{i=0}^{n-1} x_i + v \sum_{i=0}^{n-1} \max(x_i, x_{n-1-i}) \\ &= 2u \sum_{i=0}^{n-1} x_i + 2v \sum_{i=\lfloor (n+1)/2 \rfloor}^{n-1} x_i + v\delta_n x_{\lfloor n/2 \rfloor}, \end{aligned}$$

where $\delta_n = 1$ if n is odd and $\delta_n = 0$ if n is even.

Let us compute $nx_{n+1} - (n-1)x_n$ for $n \geq 2$. Let $k \geq 1$ be an integer.

- If $n = 2k$, we have

$$2kx_{2k+1} - (2k-1)x_{2k} = 2(u+v)x_{2k-1} - vx_{k-1}.$$

- If $n = 2k+1$, we have

$$(2k+1)x_{2k+2} - 2kx_{2k+1} = 2(u+v)x_{2k} - 2vx_k + vx_k = 2(u+v)x_{2k} - vx_k.$$

Collecting two cases, we have

$$\begin{aligned} nx_{n+1} - (n-1)x_n &= 2(u+v)x_{n-1} - vx_{\lfloor (n-1)/2 \rfloor} \\ \iff n(x_{n+1} - x_n) &= 2(u+v)x_{n-1} - x_n - vx_{\lfloor (n-1)/2 \rfloor} \end{aligned}$$

We have

$$x_n = \frac{2u}{n} \sum_{i=0}^{n-2} x_i + \frac{v}{n} \sum_{i=0}^{n-2} \max(x_i, x_{n-2-i}) \leq (2u+v)x_{n-2}.$$

Hence,

$$\begin{aligned} n(x_{n+1} - x_n) &\geq 2(u+v)x_{n-1} - (2u+v)x_{n-2} - vx_{\lfloor (n-1)/2 \rfloor} \\ &\geq 2(u+v)x_{n-1} - (2u+v)x_{n-1} - vx_{n-1} \\ &\geq 0. \end{aligned}$$

□

Proposition 2. *The sequence (x_n) satisfies the recurrence*

$$nx_{n+1} - (n-1)x_n = 2(u+v)x_{n-1} - vx_{\lfloor (n-1)/2 \rfloor}, \quad n \geq 1, \quad (1.3)$$

with $x_0 = x_1 = 1$.

Lemma 1. For every $n \geq 0$, we have

$$M_n = \sum_{i=0}^{n-1} \max(i^\alpha, (n-1-i)^\alpha) \leq 2 \int_{n/2}^n x^\alpha dx.$$

Proof. Let $n = 2k$. We have

$$M_{2k} = 2 \sum_{i=k}^{2k-1} i^\alpha \leq 2 \int_k^{2k} x^\alpha dx = 2 \int_{n/2}^n x^\alpha dx.$$

Let $n = 2k + 1$. We have $k^\alpha = 2 \times \frac{1}{2} \times k^\alpha \leq 2 \int_{k+1/2}^{k+1} x^\alpha dx$. Hence,

$$M_{2k+1} \leq 2 \int_{k+1/2}^{k+1} x^\alpha dx + 2 \int_{k+1}^{2k+1} x^\alpha dx = 2 \int_{k+1/2}^{2k+1} x^\alpha dx = 2 \int_{n/2}^n x^\alpha dx.$$

□

Lemma 2. For every $n \geq 1$, we have $x_n \leq Kn^\alpha$ for some $K > 0$.

Proof. We have $x_2 = 2u + v$. Assume that $x_i \leq Ki^\alpha$ for all $1 \leq i \leq n$ with $n \geq 2$ and $K > 0$ is to be determined. For $n \geq 2$, we have

$$\begin{aligned} nx_{n+1} &= 2u + 2u \sum_{i=1}^{n-1} x_i + v \sum_{i=0}^{n-1} \max(x_i, x_{n-1-i}) \\ &\leq 2u + 2uK \int_1^n x^\alpha dx + 2vK \int_{n/2}^n x^\alpha dx \\ &= 2u + \frac{2uK}{\alpha+1} (n^{\alpha+1} - 1) + \frac{2vK}{\alpha+1} (n^{\alpha+1} - (n/2)^{\alpha+1}) \\ &= 2u + \frac{Kn^{\alpha+1}}{\alpha+1} (2u + 2v(1 - 2^{-\alpha-1})) - \frac{2uK}{\alpha+1} \\ &= 2u + Kn^{\alpha+1} - \frac{2uK}{\alpha+1}. \end{aligned}$$

Note that in the sum of maxima, we do not have to isolate $\max(x_0, x_{n-1})$, since it is automatically evaluated as x_{n-1} , which is already included in the integral. Choose $K = \alpha + 1$, we check that the base cases hold

$$\begin{aligned} \alpha + 1 &\geq 1, \\ (\alpha + 1)2^\alpha &= (2(u + v) - v2^{-\alpha}) 2^\alpha = 2^{\alpha+1}(u + v) - v \geq 2u + v. \end{aligned}$$

Therefore,

$$x_{n+1} \leq Kn^\alpha \leq K(n+1)^\alpha.$$

□

Theorem 1. We have $x_n \sim Cn^\alpha$ for some $C > 0$ to be determined and α is the unique solution of the equation

$$\alpha + v2^{-\alpha} = 2(u + v) - 1$$

Proof. Let $z_n = \frac{x_n}{(n-1)^\alpha}$ for $n \geq 2$. We will prove that (z_n) converges by proving that $\sum_{n \geq 0} |z_{n+1} - z_n|$ converges. We have, for $n \geq 5$,

$$\begin{aligned} n^{\alpha+1} z_{n+1} - (n-1)^{\alpha+1} z_n &= 2(u+v)(n-2)^\alpha z_{n-1} - v \left(\left\lfloor \frac{n-1}{2} \right\rfloor - 1 \right)^\alpha z_{\lfloor (n-1)/2 \rfloor} \\ \iff z_{n+1} &= \left(1 - \frac{1}{n} \right)^{\alpha+1} z_n + \frac{2(u+v)}{n} \left(1 - \frac{2}{n} \right)^\alpha z_{n-1} - \frac{v}{n} \left(\frac{\lfloor (n-1)/2 \rfloor - 1}{n} \right)^\alpha z_{\lfloor (n-1)/2 \rfloor}. \end{aligned}$$

We have

$$\begin{aligned} \left(1 - \frac{1}{n} \right)^{\alpha+1} &= 1 - \frac{\alpha+1}{n} + O\left(\frac{1}{n^2} \right), \\ \frac{1}{n} \left(1 - \frac{2}{n} \right)^\alpha &= \frac{1}{n} + O\left(\frac{1}{n^2} \right), \\ -\frac{1}{n} \left(\frac{\lfloor (n-1)/2 \rfloor - 1}{n} \right)^\alpha &= -\frac{1}{n} \left(2^{-\alpha} + O\left(\frac{1}{n} \right) \right) = -\frac{2^{-\alpha}}{n} + O\left(\frac{1}{n^2} \right). \end{aligned}$$

By the previous lemma, (z_n) is bounded. Therefore,

$$\begin{aligned} z_{n+1} - z_n &= -\frac{\alpha+1}{n} z_n + \frac{2(u+v)}{n} z_{n-1} - \frac{v2^{-\alpha}}{n} z_{\lfloor (n-1)/2 \rfloor} + O\left(\frac{1}{n^2} \right) \\ &= -\frac{2(u+v)}{n} (z_n - z_{n-1}) + \frac{v2^{-\alpha}}{n} (z_n - z_{\lfloor (n-1)/2 \rfloor}) + O\left(\frac{1}{n^2} \right). \end{aligned}$$

Let $\nabla z_n = z_n - z_{n-1}$. Then, there exists a constant $L > 0$ such that for every $n \geq 5$,

$$|\nabla z_{n+1}| \leq \frac{2(u+v)}{n} |\nabla z_n| + \frac{v2^{-\alpha}}{n} \sum_{i=\lfloor (n+1)/2 \rfloor}^n |\nabla z_i| + \frac{L}{(n+1)^2}.$$

Equivalently, for every $n \geq 6$,

$$|\nabla z_n| \leq \frac{2(u+v)}{n-1} |\nabla z_{n-1}| + \frac{v2^{-\alpha}}{n-1} \sum_{i=\lfloor n/2 \rfloor}^{n-1} |\nabla z_i| + \frac{L}{n^2}.$$

Let $u_m = \max_{2^m \leq n < 2^{m+1}} |\nabla z_n|$. Then, for every $n \in [2^m, 2^{m+1})$, we have

$$\begin{aligned} \frac{1}{n-1} |\nabla z_{n-1}| &\leq \frac{1}{n-1} \max_{2^{m-1} \leq i < 2^m} |\nabla z_i| = \frac{1}{n-1} \max(u_{m-1}, u_m), \\ \frac{1}{n-1} \sum_{i=\lfloor n/2 \rfloor}^{n-1} |\nabla z_i| &\leq \frac{n+1}{2(n-1)} \max_{2^{m-1} \leq i < 2^m} |\nabla z_i| \leq \left(\frac{1}{2} + \frac{1}{n-1} \right) \max(u_{m-1}, u_m). \end{aligned}$$

Therefore,

$$\begin{aligned} |\nabla z_n| &\leq \frac{2(u+v)}{n-1} \max(u_{m-1}, u_m) + v2^{-\alpha} \left(\frac{1}{2} + \frac{1}{n-1} \right) \max(u_{m-1}, u_m) + \frac{L}{2^{2m}} \\ &= \left(\frac{\alpha+1}{n-1} + v2^{-\alpha-1} \right) \max(u_{m-1}, u_m) + \frac{L}{2^{2m}}. \end{aligned}$$

Since $v2^{-\alpha} = 2(u+v) - \alpha - 1 < 1$, we can choose m_0 large enough such that for every $m \geq m_0$ and $n \in [2^m, 2^{m+1})$, there exists γ_1 such that $\frac{\alpha+1}{n-1} + v2^{-\alpha-1} < \gamma_1 < \frac{1}{2}$. Taking the maximum over $n \in [2^m, 2^{m+1})$, we have

$$u_m \leq \gamma_1 \max(u_{m-1}, u_m) + \frac{L}{2^{2m}}.$$

If $u_{m-1} > u_m$, then

$$u_m \leq \gamma_1 u_{m-1} + \frac{L}{2^{2m}} < \gamma_1 u_{m-1} + \frac{L}{2^{2m}(1-\gamma_1)}.$$

If $u_{m-1} \leq u_m$, then

$$u_m \leq \frac{L}{2^{2m}(1-\gamma_1)} < \gamma_1 u_{m-1} + \frac{L}{2^{2m}(1-\gamma_1)}.$$

In both cases, we have

$$\begin{aligned} u_m &\leq \gamma_1 u_{m-1} + \frac{L}{2^{2m}(1-\gamma_1)} \\ &\leq \sum_{i=m_0+1}^m \gamma_1^{m-i} \frac{L}{2^{2i}(1-\gamma_1)} + \gamma_1^{m-m_0} u_{m_0} \\ &< \frac{L}{(1-\gamma_1)} \sum_{i=m_0+1}^m \gamma_1^{m-i} \left(\frac{1}{4}\right)^i + \gamma_1^{m-m_0} u_{m_0}. \end{aligned}$$

Choose γ_2 such that $\max\left(\gamma_1, \frac{1}{4}\right) < \gamma_2 < \frac{1}{2}$. We have

$$u_m < \frac{L}{(1-\gamma_1)} \sum_{i=m_0+1}^m \gamma_2^m + \gamma_2^{m-m_0} u_{m_0} \leq L' m \gamma_2^m,$$

for some constant $L' > 0$. Choose γ_3 such that $\gamma_2 < \gamma_3 < \frac{1}{2}$. Then there exists $m_1 > m_0$ such that for every $m \geq m_1$, we have $L' m \gamma_2^m < L' \gamma_3^m$, implying that,

$$u_m < L' \gamma_3^m.$$

Therefore,

$$\sum_{n=0}^{\infty} |\nabla z_n| = \sum_{m=0}^{\infty} \sum_{n=2^m}^{2^{m+1}-1} |\nabla z_n| = \sum_{m=0}^{\infty} 2^m u_m < \sum_{m=0}^{m_1-1} 2^m u_m + L' \sum_{m=m_1}^{\infty} (2\gamma_3)^m < \infty.$$

Hence, (z_n) converges to some constant $C > 0$. Thus, $x_n \sim Cn^\alpha$. □